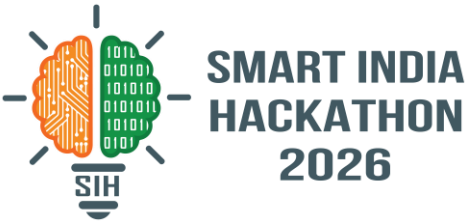


- [illegible]



D-CRYPT SHIELD

India's Blockchain Intelligence & VASP Attribution Engine



❑ Main Problem

Criminals route stolen **crypto** through anonymous **unhosted wallets, mixers, and multi-chain hops** to conceal the destination. Because police can only issue legal freeze requests via the SAHYOG portal to **known exchanges, slow manual tracing** creates critical delays that let criminals cash out before assets can be frozen. [Deep-dive documentation](#)

❑ Core Issues

⚠ Multi-Hop Cross-Chain Tracing
Funds move through intermediary wallets, bridges, mixers and multiple chains, making manual tracing difficult.

⚠ Raw Blockchain Data
Thousands of transactions and wallet addresses are difficult to interpret without an investigation-focused graph.

⚠ Unknown VASP
Investigators may identify the suspect wallet but not the VASP receiving the funds.

⚠ Delayed Action
VASP attribution, evidence preparation and lawful request routing require additional manual effort.

⚖️ ❑ Our Solutions

🛡 Omni-Chain BFS Engine
Traces cross-chain jumps via Price Oracles (BTC/ETH+Multi Chains); flags probable mixer/tumbler usage and halts automated tracing there for investigator-led OSINT follow-up.

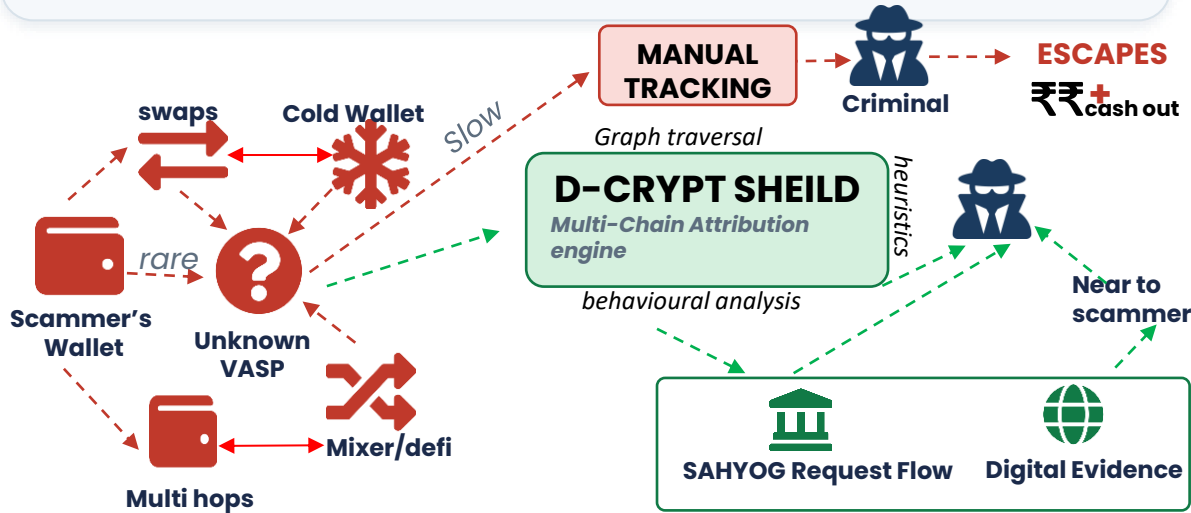
🛡 Visual Intelligence
Real-time interactive Graph Visualization & Portfolio Overview of the network.

🛡 AI VASP & Threat Scoring
XGBoost model scores wallets 0-100 using deposit-pattern regularity, hop count to nearest tagged VASP, transaction timing correlation, and in/out-degree driving lawful asset-freezing requests.

🛡 Automation + Sahyog API integration
Dual onboarding (civilian + Sahyog); auto-generates Sec 91 notices with Sec 65B QR codes.

❑ Our Unique Innovations

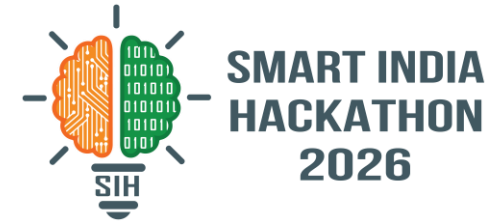
- **Multi-Chain Tracing**
Trace transactions across supported blockchain networks.
- **VASP Attribution**
Identifies the nearest deposit-accepting VASP via BFS hop-count to tagged deposit addresses, combined with co-spend clustering and known-address-database matching.
- **Graph Intelligence**
Reconstruct wallet-to-wallet fund movement visually
- **Investigation Workflow**
Generate evidence-backed reports and prepare lawful requests for SAHYOG.
- **Active Intelligence**
OSINT correlation (Telegram/X) feed as auxiliary signals into the Threat Score, strengthening attribution confidence alongside on-chain analysis.





TECHNICAL APPROACH

METHODOLOGY



INPUT

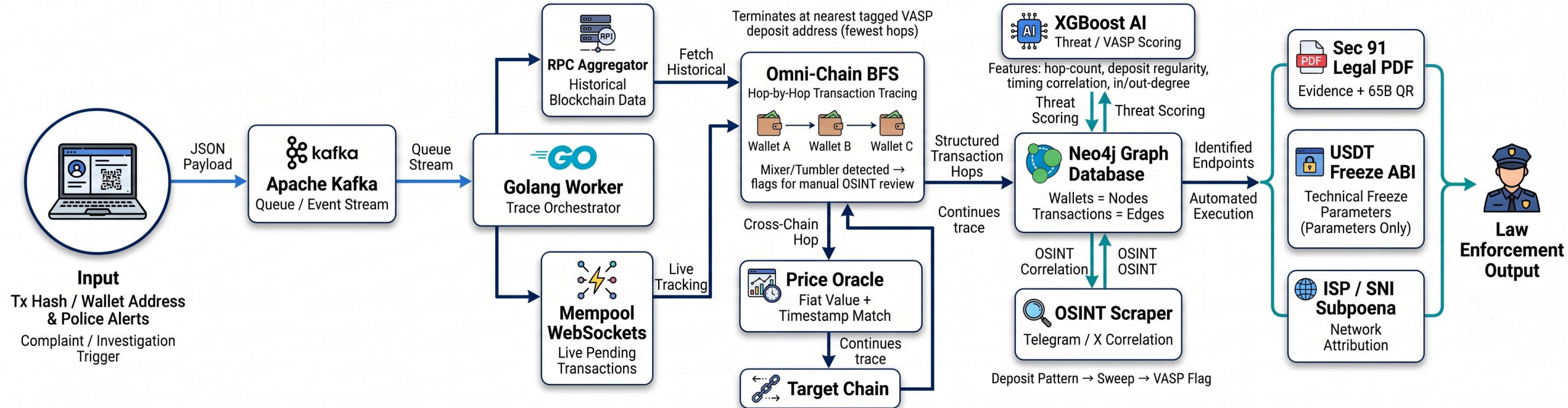
INGESTION

CENTER EXECUTION & BLOCKCHAIN TRACE ENGINE

BLOCKCHAIN TRACE / CROSS-CHAIN PROCESSING

INTELLIGENCE + GRAPH DATABASE

LEGAL & LAW-ENFORCEMENT OUTPUT



Backend Core



Golang · Geth/Ethers.js · Custom RPC Aggregator

Intelligence Service



Python · XGBoost · OSINT Scrapers · ReportLab/PyQRCode

Frontend Dashboard



Next.js · React · Tailwind CSS · D3.js

Database Engine



Neo4j Graph · PostgreSQL

DevOps & Streaming



Docker · Kubernetes · Apache Kafka



FEASIBILITY AND VIABILITY



**Data Feasibility**

Blockchains are public, ledgers extraction is available via standard RPC nodes.

**Financial Feasibility**

Open-source core with pluggable RPC/indexing providers for civilian-tier lookups; self-hosted archive nodes for classified/investigation-tier tracing to avoid third-party IP exposure.

**Legal Feasibility**

Designed to support existing lawful disclosure workflows with evidence provenance, cryptographic integrity checks and investigator review.

**Operational Feasibility**

API-ready architecture for integration with SAHYOG and case-based LEA workflows.

Potential Challenge	Strategy for Overcoming It
API Rate Limiting — Tracing 100k-node networks triggers free-tier RPC IP blocks.	RPC Aggregator — Asynchronous queueing + RPC provider failover + rate-limit aware workers
DB Memory Exhaustion — Neo4j's RAM-intensive multi-hop queries risk OOM crashes.	K8s & Query Pagination — Query pagination + bounded graph traversal + horizontal scaling
Data Sovereignty — MHA cannot host classified investigations on public cloud	Air-Gapped Deployment — Self-hosted nodes and sovereign archive infrastructure for classified cases, eliminating reliance on third-party RPC providers that log requester IPs.
Legal Admissibility — AI-traced PDFs risk rejection if tampering is alleged in court.	Sec 65B Crypto QR — Cryptographic hashes + timestamps + provenance metadata support evidence integrity and verification.
Stagnant / Offshore Funds — Funds never move, or foreign VASPs refuse IP request	Limited/ Offshore VASP Cooperation — Provides evidence package and jurisdiction-aware request data for authorized disclosure or freezing workflows.



IMPACT AND BENEFITS



CORE BENEFICIARIES



Civilians (Victims)

Instant SAHYOG tracing freezes stolen funds before the scammer can cash out.



State Police (LEAs)

Investigator-Assisted Automation. Multi-Day Manual Tracing → Automated Analysis



MHA & Government

100% data sovereignty. Replaces expensive foreign tracking tools while keeping intelligence in India.

LEGAL & COURTROOM IMPACT



Evidence Ready Investigation

Cryptographic hashes, timestamp that digital evidence cannot be fully ignored by defense lawyers.



Automated Jurisdiction

Generates jurisdiction-aware disclosure/freeze request packages.



Conviction Engine

Shifts police focus away from tedious manual tracing and directly toward securing legal convictions.



Plug-and-Play Sovereign Hub

MHA centrally hosts the core, granting instant API access to every State Police department.



ACTIONABLE VSAP ATTRIBUTION

Unknown Wallet → Probable VASP → Confidence Score → Evidence Package



The Tri-Factor Benefits

Faster Investigation

Automates repetitive blockchain tracing and reduces the time required to identify probable VASPs.

Stronger Cyber Crime Response

Helps LEAs follow illicit crypto flows, identify laundering patterns and act faster on actionable intelligence.

Sovereign & Scalable

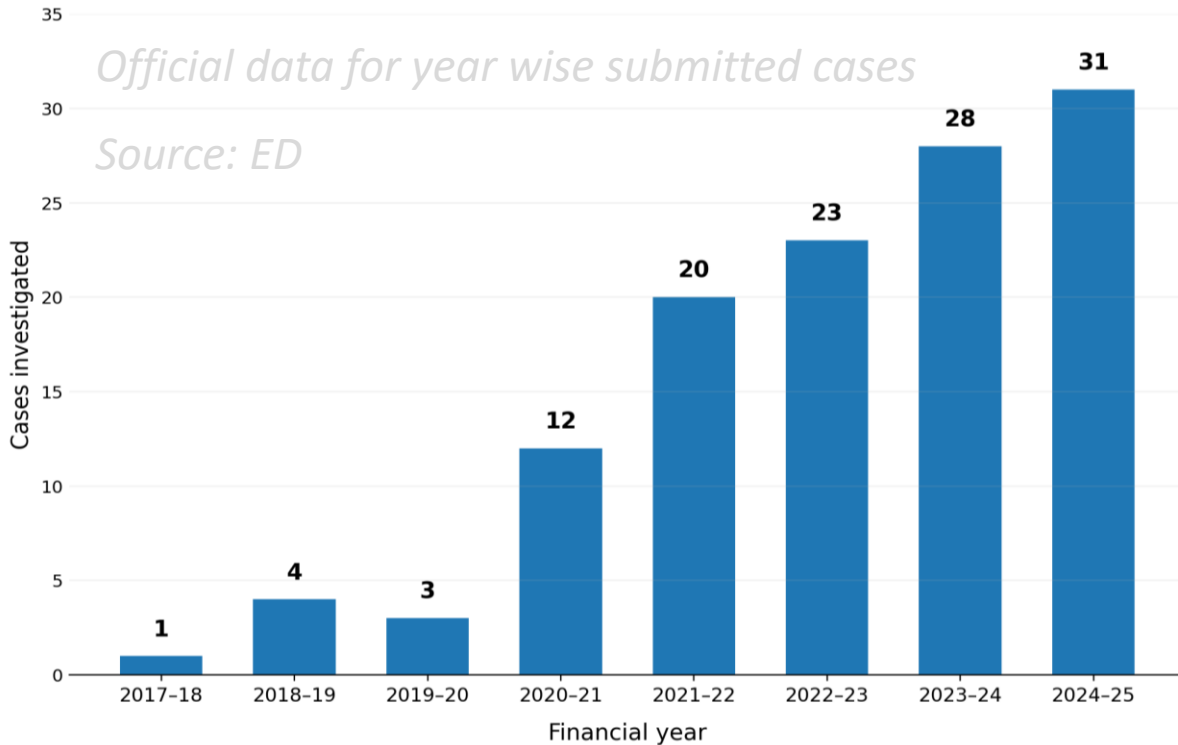
Designed for controlled government deployment with scalable multi-chain analysis and data ownership.



RESEARCH AND REFERENCES



❑ Problem Validation



📶 **RPC IP Logging** [click](#)
CoinDesk: Consensys (Nov 2022) confirms Infura logs physical IPs.

❑ Project Related Links

Video



[Prototype Explanation](#)

Prototype



shield.d-crypt.in

PDF



[Explore full solution doc](#)

❑ Technical & Legal Execution Proofs

📈 **AI Graph Science** [click](#)
Neo4j academic docs for Heuristic Graph Clustering in Financial Fraud.

🔑 **Legal Evidence** [click](#)
CrPC (Sec 91) & Indian Evidence Act (Sec 65B) legal frameworks.

❄️ **USDT Freezing** [click](#)
Verified Etherscan USDT contract addBlackList ABI used for LE freezes.

📶 **Pre-Emptive Tracking** [click](#)
Official Web3.js docs for eth.subscribe('pendingTransactions').